

NexusFinance — API & Database Reference

Digital Lending Platform for Cambodia

EN

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1. API Overview

Base URL: <https://nexusfinance-lof3.onrender.com/api>

All endpoints return JSON. Protected endpoints require a Bearer token in the Authorization header.

Status Code	Meaning
200	Success
400	Bad request (missing/invalid fields)
401	Unauthorized (invalid or missing token)
403	Forbidden (insufficient role)
404	Resource not found
429	Rate limited
500	Internal server error

2. Authentication Endpoints

POST /api/auth/register

Register a new user account.

Field	Type	Required	Description
name	string	Yes	User's full name
email	string	Yes	Valid email address
password	string	Yes	Min 6 characters
phone	string	No	Phone number

Response: 201 { message: 'Account created. Please verify your email.' }

POST /api/auth/login

Authenticate and receive a JWT token.

Field	Type	Required	Description
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email	string	Yes	Registered email
password	string	Yes	Account password

Response: 200 { token: 'jwt...', user: { id, name, email, role } }

Response (unverified email): 403 { error: '...', code: 'EMAIL_NOT_VERIFIED' } — verify your email via send-otp + verify-otp first.

POST /api/auth/send-otp

Send a 6-digit OTP to the user's email.

Field	Type	Required	Description
email	string	Yes	Email to send OTP to

POST /api/auth/verify-otp

Verify the OTP code.

Field	Type	Required	Description
email	string	Yes	Email address
code	string	Yes	6-digit code

3. Loan Endpoints

GET /api/loans

List all loans. Customers see only their own; admins see all.

Response: 200 [{ id, user_id, name, email, amount, type, status, ... }]

POST /api/loans

Create a new loan application. Requires authentication.

Field	Type	Required
name	string	Yes

email	string	Yes
amount	number	Yes
type	string	Yes
purpose	string	No
monthlyIncome	string	No
durationMonths	string	No

PATCH /api/loans/:id

Update a loan status. Requires loan-officer or super-admin role.

Field	Type	Required
status	string	Yes (approved/rejected/ hold)

Dedicated actions: PATCH /api/loans/:id/approve, /api/loans/:id/reject, /api/loans/:id/hold — all restricted to loan-officer and super-admin.

4. Transaction Endpoints

GET /api/transactions

List transactions. Customers see their own; admins see all.

POST /api/transactions

Record a new transaction.

POST /api/simulate-payment

Simulate a payment callback (for testing). Triggers Telegram notification.

5. Admin Endpoints

The following endpoints require a loan-officer or super-admin role:

Endpoint	Method	Description
/api/stats	GET	Platform statistics
/api/users	GET	List all users (paginated)
/api/audit/logs	GET	Audit log entries
/api/config	GET/PATCH	Platform configuration
/api/tasks	GET	KYC/review tasks

6. KHQR Endpoints

Endpoint	Method	Description
/api/khqr/generate	POST	Generate a KHQR code
/api/khqr/verify	POST	Verify a payment by MD5
/api/khqr/decode	POST	Decode a KHQR string
/api/khqr/deeplink	POST	Generate a payment deeplink

7. PayWay Endpoints

NexusFinance uses the ABA PayWay Purchase API with hosted checkout. The frontend calls `/api/payway/purchase`, then submits the returned fields as a form to ABA's checkout URL. The return `/cancel` endpoints handle the post-payment redirects.

Endpoint	Method	Description
/api/payway/purchase	POST	Create a hosted-checkout purchase (view_type=hosted_view) and persist a PENDING record
/api/payway/return	GET	ABA success redirect; verifies status, records approved payments, redirects to <code>/payment/success</code>
/api/payway/cancel	GET	ABA cancel redirect; redirects to <code>/payment/cancel</code>
/api/payway/verify-payment	POST	Check transaction status; marks APPROVED + writes ledger
/api/payway/callback	POST	ABA webhook (HMAC-signed) with approval/decline updates

/api/payway/simulate-payment	POST	Sandbox-only helper to approve a payment for testing
/api/payway/transactions	GET	List recent PayWay transactions from the database
/api/payway/generate-qr	POST	Legacy endpoint; now delegates to the Purchase API

Approved PayWay payments are written to nexus_payway_transactions and create a Repayment entry in nexus_transactions, plus an in-app and Telegram notification.

Purchase requests are signed with HMAC-SHA512 over a fixed 24-field hash order (req_time, merchant_id, tran_id, amount, items, shipping, firstname, lastname, email, phone, type, payment_option, return_url, cancel_url, continue_success_url, return_deeplink, currency, custom_fields, return_params, payout, lifetime, additional_params, google_pay_token, skip_success_page). view_type and payment_gate are sent unsigned. Webhook callbacks are verified with the same hash order via the x-payway-hmac-sha512 header.

8. Database Schema Detail

nexus_users

Column	Type	Description
id	SERIAL PK	Auto-increment ID
name	TEXT	User's full name
email	TEXT UNIQUE	Email address
password	TEXT	Bcrypt password hash
role	TEXT	customer / loan-officer / super-admin
phone	TEXT	Phone number
telegram_chat_id	BIGINT	Telegram chat ID (if linked)
email_verified	BOOLEAN	Whether the email has been verified (blocks login when false)
otp_code	TEXT	Bcrypt hash of OTP
otp_expires_at	TIMESTAMPTZ	OTP expiry time
otp_verified_at	TIMESTAMPTZ	When OTP was verified
created_at	TIMESTAMPTZ	Account creation time

nexus_loans

Column	Type	Description
id	SERIAL PK	Auto-increment ID
user_id	INTEGER	Applicant user ID
name	TEXT	Applicant name (denormalized)
email	TEXT	Applicant email (denormalized)
amount	NUMERIC	Loan amount
type	TEXT	Loan type (SME, Personal, etc.)
status	TEXT	pending / approved / rejected / hold
purpose	TEXT	Loan purpose
duration_months	INTEGER	Repayment duration
interest_rate	NUMERIC	Annual interest rate
monthly_income	NUMERIC	Applicant monthly income
assigned_to	INTEGER	Assigned officer user ID
created_at	TIMESTAMPTZ	Application date

nexus_payway_transactions

Column	Type	Description
id	BIGSERIAL PK	Auto-increment ID
tran_id	TEXT UNIQUE	PayWay transaction ID
email	TEXT	Customer email (normalized)
amount	NUMERIC	Payment amount
currency	TEXT	Currency (default USD)
status	TEXT	PENDING / APPROVED / DECLINED
apv	TEXT	PayWay approval code

loan_id	TEXT	Linked loan ID (optional)
user_id	INT	NexusFinance user ID
created_at	TIMESTAMPTZ	Creation time
paid_at	TIMESTAMPTZ	When the payment was approved